Jin-Chuan Shi

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Research Interests

My research lies at the intersection of 3D reconstruction, dexterous manipulation, and large-scale machine learning. I work on reconstructing hand–object interaction from in-the-wild video and retargeting it into physically consistent simulation, turning Internet-scale human videos into robot-usable manipulation data for general-purpose dexterous manipulation.

Core research questions: (1) How can we build scalable, efficient, and robust reconstruction-and-retargeting systems that turn massive Internet videos into high-quality manipulation data? (2) On top of such data, how can we build scalable robot learning systems that unify perception and dexterous manipulation, enabling reliable general-purpose robots in unstructured, real-world environments?

Education

College of Computer Science and Technology, Zhejiang University — **Hangzhou, China**
Ph.D. in Computer Science and Technology — *Sept. 2025 – Mar. 2029 (expected)*

- Advisors: Prof. Chunhua Shen, Prof. Hao Chen

School of Computer Science and Engineering, Beihang University — **Beijing, China**
M.Eng. in Computer Science and Technology — *Sept. 2022 – Jan. 2025*

- Advisors: Prof. Shi-Min Hu (Tsinghua University), Prof. Miao Wang

Shen Yuan Honors College, Beihang University — **Beijing, China**
B.Eng. in Computer Science and Technology — *Sept. 2018 – June 2022*

- Advisor: Prof. Miao Wang

Publications

* Equal contribution. Author of this CV in **bold**.

Preprints / Under Review

[P1] Chenhao Bai, Liqin Lu, Kaijun Wang, Hui Chen, **Jin-Chuan Shi**, Yuyang Liu, Hao Chen, Chunhua Shen. HORIZON: Recoverability-Governed Curriculum for Physical-Domain Scaling. Under review at *Conference on Robot Learning (CoRL)*, 2026.

[P2] Weijie Wang*, Zimu Li*, **Jin-Chuan Shi**, Zeyu Zhang, Botao Ye, Marc Pollefeys, Donny Y. Chen, Bohan Zhuang. TriSplat: Simulation-Ready Feed-Forward 3D Scene Reconstruction. Under review at *ACM SIGGRAPH Asia*, 2026.

[P3] **Jin-Chuan Shi***, Chengye Su*, Jiajun Wang, Ariel Shamir, Miao Wang. Mono4DEditor: Text-Driven 4D Scene Editing from Monocular Video via Point-Level Localization of Language-Embedded Gaussians. Under review at *Computational Visual Media Journal (CVMJ)*, 2026.

Peer-Reviewed Publications

[C1] **Jin-Chuan Shi***, Binhong Ye*, Tao Liu, Xiaoyang Liu, Yangjinhui Xu, Junzhe He, Zeju Li, Hao Chen, Chunhua Shen. AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation. *ACM SIGGRAPH (Conference Track)*. 2026.

[C2] Bowen Yang, Zishuo Li, Yang Sun, Changtao Miao, Yifan Yang, Man Luo, Xiaotong Yan, Feng Jiang, **Jin-Chuan Shi**, Yankai Fu, Ning Chen, Junkai Zhao, Pengwei Wang, Guocai Yao, Shanghang Zhang, Hao Chen, Zhe Li, Kai Zhu. AoE: Always-on Egocentric Human Video Collection for Embodied AI. *CVPR Workshop on Embodied AI in the Era of Large Models*. 2026.

[C3] **Jin-Chuan Shi**, Miao Wang, Hao-Bin Duan, Shao-Hua Guan. LEGaussians: Language Embedded 3D Gaussians for Open-Vocabulary Scene Understanding. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. 2025. **268 citations** (Google Scholar, Jul. 2026).

[C4] Hao-Bin Duan, Miao Wang, **Jin-Chuan Shi**, Xu-Chuan Chen, Yan-Pei Cao. BakedAvatar: Baking Neural Fields for Real-Time Head Avatar Synthesis. *ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia)*. 2023.

[C5] Miao Wang, Yi-Jun Li, **Jin-Chuan Shi**, Frank Steinicke. SceneFusion: Room-Scale Environmental Fusion for Efficient Traveling Between Separate Virtual Environments. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*. 2023.

[C6] Yi-Jun Li, **Jin-Chuan Shi**, Fang-Lue Zhang, Miao Wang. Bullet Comments for 360° Video. *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 2022.

[C7] Miao Wang, Ziming Ye, **Jin-Chuan Shi**, Yong-Liang Yang. Scene-Context-Aware Indoor Object Selection and Movement in VR. *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 2021.

Honors and Awards

- National Scholarship, China Ministry of Education, 2024
- Outstanding Graduate Student, Beihang University, 2024
- Outstanding Graduates of Beijing's Higher Education Institutions, 2022
- Freshman Academic Scholarship (Special Prize), Beihang University, 2018
- Outstanding Academic Scholarship, Beihang University, 2019–2021, 2023
- Outstanding Student Leader Award, Beihang University, 2019–2021

Service

Reviewer: ACM SIGGRAPH (Asia), NeurIPS, Pacific Graphics (PG), IEEE ISMAR

Skills

Programming: Python, PyTorch, C, C++, CUDA, C#, Unity3D, LaTeX, Linux, JavaScript, Java
Research areas: 3D Reconstruction, Neural Rendering, Gaussian Splatting, Hand-Object Interaction, Dexterous Manipulation, VR/AR
Languages: English (fluent, CET-4 & CET-6), Mandarin (native)